ACCESS : A Benchmark for Abstract Causal Event Discovery and Reasoning

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Abstract

Identifying cause-and-effect relationships is critical to understanding real-world dynamics and ultimately causal reasoning. Existing methods for identifying event causality in NLP, including those based on Large Language Models (LLMs), exhibit difficulties in out-of-distribution settings due to the limited scale and heavy reliance on lexical cues within available benchmarks. Modern benchmarks, inspired by probabilistic causal inference, have attempted to construct causal graphs of events as a robust representation of causal knowledge, where CRAB (Romanou et al., 2023) is one such recent benchmark along this line. In this paper, we introduce ACCESS, a benchmark designed for discovery and reasoning over abstract causal events. Unlike existing resources, ACCESS focuses on causality of everyday life events on the abstraction level. We propose a pipeline for identifying abstractions for event generalizations from GLUCOSE (Mostafazadeh et al., 2020), a large-scale dataset of implicit commonsense causal knowledge, from which we subsequently extract $1,4$K causal pairs. Our experiments highlight the ongoing challenges of using statistical methods and/or LLMs for automatic abstraction identification and causal discovery in NLP. Nonetheless, we demonstrate that the abstract causal knowledge provided in ACCESS can be leveraged for enhancing QA reasoning performance in LLMs.

1 Introduction

Commonsense causal reasoning plays a vital role in developing a mental model of reality, where the ability to discover, explain and predict causal relations between events or forces in the environment is fundamental to human planning and control (Johnson-Laird and Khemlani, 2017; Griffiths, 2017). Cognitive science studies further suggest that event causality is critical to human understanding of narratives (Van den Broek et al., 1996; Fletcher and Bloom, 1988; Tillman et al., 2020; Sun et al., 2023), and story events with more causal relations tend to be better memorized than those with fewer relations (Graesser et al., 2003). Humans are able to construct a causal mental model of events after reading a set of stories (Zwaan et al., 1995). For example, in Figure 1, a reader would easily identify a causal relation between $e1$: "A person needs money." and $e2$: "A person gets a job." by *abstracting* away concrete details, such as mentions of particular entities, grouping linguistic variations of the same meanings, and observing that $e1$ almost always leads to $e2$ in multiple stories, without explicit presence of lexical cues (e.g. *because*) in text. Thus, this paper focuses on investigating to what extent LLMs can identify causal relations *without relying on linguistic cues* and perform causal reasoning over commonsense knowledge on the abstraction level.

Prior works on *causal relation extraction* heavily rely on linguistic cues, e.g. *because of, by, due to*, to discern causal relations between event mentions and cause/effect text spans within a text (Wolff and Song, 2003; Mirza and Tonelli, 2014). In contrast, statistical *causal discovery* methods for event causality do not require linguistic cues but exploit statistical information of symbolic representations of events (Pearl and Mackenzie, 2018). As a result, those approaches are able to find causal relations even when they are not explicitly mentioned anywhere in texts. Therefore, there has been criticism regarding the susceptibility of these causal relation extraction models to exploit the linguistic cues to attain high performance without engaging in actual causal reasoning (Yang et al., 2022; Li et al., 2022). Ample of causal relation extraction datasets, including TempEval-3 (Mirza et al., 2014), CATENA (Mirza and Tonelli, 2016), Causal-TimeBank (Mirza and Tonelli, 2014), BECauSE (Dunietz et al., 2015, 2017) and Event StoryLine Corpus (Caselli and Vossen, 2017), are not suitable for evaluating statis-

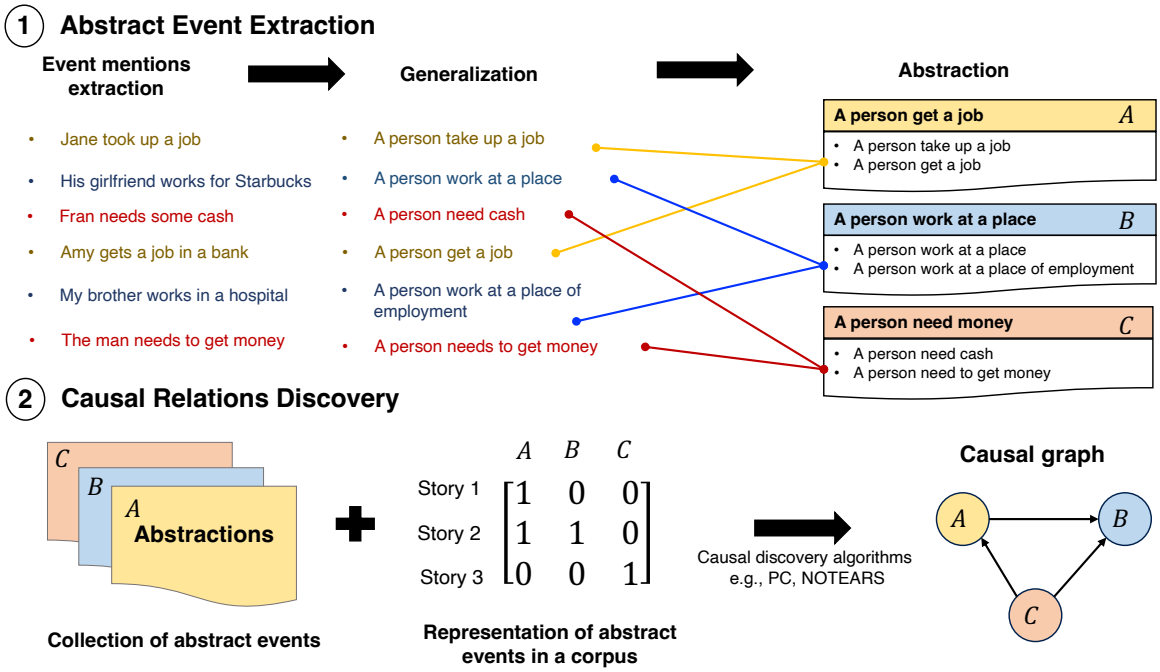

Figure 1: Pipeline of abstract causal event discovery. An event is viewed from three hierarchical levels: **mention** (realization in a specific text corpus), **generalization** (conceptualization of the event's components) and **abstraction** (group of causally consistent generalizations). Given a collection of event mentions, Phase 1 produces a collection of abstractions A, B, C that are mapped back to the original corpus to construct a suitable representation in Phase 2, such as a co-occurrence matrix. Causal discovery algorithms can then be employed to detect causal relations within the data, which may consider the contexts.

tical causal discovery methods, because mentions of causal events that are semantically similar but expressed in different linguistic forms are not mapped into the same symbols.

Humans identify grouping of semantic similar event mentions via *abstraction*, which omits concrete spurious details. It has also been suggested that effective causal reasoning requires models to learn suitable abstract representation of the world (Girju, 2003). Abstraction of events can well leverage causality theories (Pearl, 2009), which provide a theoretical underpinning for causal reasoning and formal analysis of causal relations. Given a set of random variables, theoretically grounded causal discovery algorithms (Vowels et al., 2022) use statistics collected from a dataset to construct *causal graphs*, which combine causal relations into logically coherent directed acyclic graphs (DAGs). In a causal graph, a node v_i denotes a random variable, while an edge from v_i to v_j indicates v_i is a direct cause of v_j. To the best of our knowledge, none of the existing datasets, such as COPA (Roemmele et al., 2011) and MAVEN-ERE (Wang et al., 2022), provide both abstraction of events grounded in a corpus and the corresponding causal graphs.

Existing knowledge graphs containing causal relations (Sap et al., 2019; Hwang et al., 2021; Hassanzadeh et al., 2022; Mbouadeu et al., 2023), cannot support evaluating both (1) event abstractions grounded in a corpus, and (2) construction of commonsense causal graphs from a collection of documents. The popular ATOMIC (Sap et al., 2019) in particular is harvested by crowd-sourcing so that similar events in ATOMIC are not grouped together and there is also no associated corpus containing all relevant event mentions. As a result, it is challenging to recover the grounded contexts and map events to random variables in order to apply statistical causal discovery. Two additional resources for constructing causal graphs from a collection of documents are CauseNet (Heindorf et al., 2020) and CRAB (Romanou et al., 2023). The causal relations from CauseNet are explicitly mentioned in texts while causal semantics can exist beyond lexical mentions. CRAB generates causal graphs from events extracted from online news articles. Neither CauseNet or CRAB provides abstraction of events or grouping of semantically equivalent events with linguistic variations. Therefore, the resulting graphs over fine-grained events can explode in size, posing

severe computational difficulties. Another related resource is GLUCOSE (Mostafazadeh et al., 2020), which translates natural language expressions of event mentions and their relations into generic inferential rules dependent on story contexts. In those rules, entities are mapped to generalized concepts and the same keywords are used for the same relations, such as *causes*. Although GLUCOSE lacks grouping of similar abstract events and the interconnection of relations into a logically coherent causal graph, our paper extends this dataset to construct abstract knowledge graphs of causality.

Contribution. We introduce **ACCESS**, a benchmark for **A**bstra**C**t **C**ausal **E**vent Di**S**covery and Rea**S**oning. We propose to explore event causality at the abstraction level as a more efficient representation of knowledge. We introduce a reusable pipeline (See Figure 1) to curate causal relations from a large corpus of stories of daily life events. The resulting benchmark, ACCESS, is a graphical modelling of causal relations of 725 event abstractions forming a base of abstract commonsense knowledge of causation. Our benchmark also includes annotations to evaluate each step of the pipeline. Using ACCESS, our experiments shed light on the ongoing challenges within the field.

Firstly, the application of statistical structure learning algorithms for full graph discovery remains highly challenging. Secondly, solely relying on LLMs and automatic clustering proves insufficient for adequate event abstraction. Thirdly, LLMs still struggle with pairwise non-contextual causal discovery, indicating a gap in their possession of complete commonsense causal knowledge. Lastly, incorporating abstract commonsense knowledge through a causal graph enhances Question Answering (QA) reasoning tasks in LLMs up to 20%.

2 Causal Event Abstraction

We follow the definition of events provided in TimeML (Pustejovsky et al., 2003), ECB+ Annotation Guidelines (Cybulska and Vossen, 2014) and Event StoryLine Corpus (Caselli and Vossen, 2017). An **event** refers to any situation or state that happens or holds, which consists of four basic components: `action/state`, `location`, `time` and `participant(s)`. We here consider `location` and `time` as optional; for instance, the sentence *he goes to sleep* is sufficiently an event. Each component of an event is associated with a concept in an ontology.[1] A realization of a concept in the event is an **event mention**. An **event abstraction** is a tuple ⟨`action (state)/concept`, `participant/concept`, `time/concept`, `location/concept`⟩ shared among all mentions of that event, where each component is either an entity or a concept at an appropriate abstraction level. An event abstraction is itself an event and can be identified by replacing every component in its representation by a more abstract concept in the ontology. For example, *His girlfriend* `[person]` *works* `[action]` *for Starbucks* `[location]` *on the weekends* `[time]`.

From another point of view, an event abstraction is a generalization of a **cluster** of event mentions that describe the same event. Two event mentions are **equivalent** if they are associated with the same event abstraction. An event abstraction is **causally consistent** w.r.t. a set of event mentions, if (1) none of its mention pairs at the semantic level contains a causal relationship, and (2) the semantics of all its mentions are either the cause or the effect of mentions in another event abstraction. Table 1 describes all the terms used in this paper and throughout the annotation process.

Definition of causation. Based on the counterfactual theory of causation (Lewis, 2013), an event x is said to **cause** another event y and event y is said to be an **effect** of event x if (1) event y temporally follows event x directly i.e., there are no intermediate events or if there is one, it must rarely occur, and (2) event y would not commonly occur if event x did not occur. It is worth noting that unlike such datasets as BECauSE (Dunietz et al., 2017) or CauseNet (Heindorf et al., 2020) that consider causality between concepts, here causality is defined on the event (sentence) level, which takes into account the interaction of multiple participants. In statistical causality literature, there exist 3 causal structures of interest: *confounder*, *collider* and *mediator*. For random variables X, Y, Z,

- Z is a called confounder if it causes both X and Y, written as $X \leftarrow Z \rightarrow Y$;
- Z is a collider when Z is a common child of X and Y but X and Y themselves are not related, written as $X \rightarrow Z \leftarrow Y$;

[1] Ontology refers to a collection of concepts and their relations within a domain (Gruber, 1993).